Topic: Section 1

Place your screenshot of your ultrasonic sensor data. Add a brief caption (1-3 sentences) that describes the pros and cons of the ultrasonic sensor, as well as your approximate sampling rate and how you are cleaning out bad data.

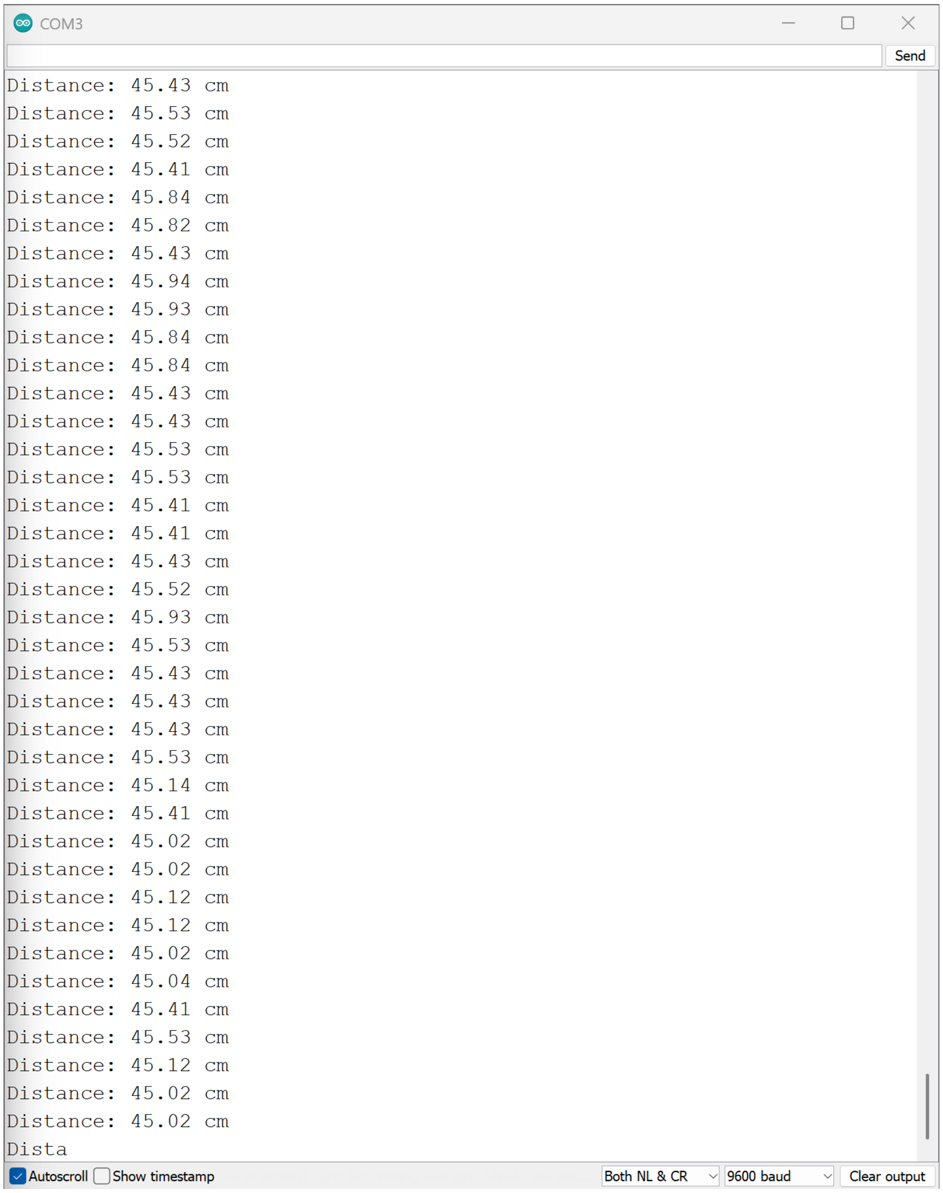

Figure 1: monitorData

Some pros would be that they are really cheap and availible everywhere, some cons would be that there is a limit to the material that can be sensed by them like soft materials. My fitering was very pretty, I pretty much just went 2cm<x<400cm and everything that was outside of that I didnt accept sice the sensors specs were set there. The aproximate sampling was found by working back from the distance limits of the sensor and finding what the frequency would be to match that. I didnt do anything fancy like with a type of dynamic smaple frequency. That would be cool though.

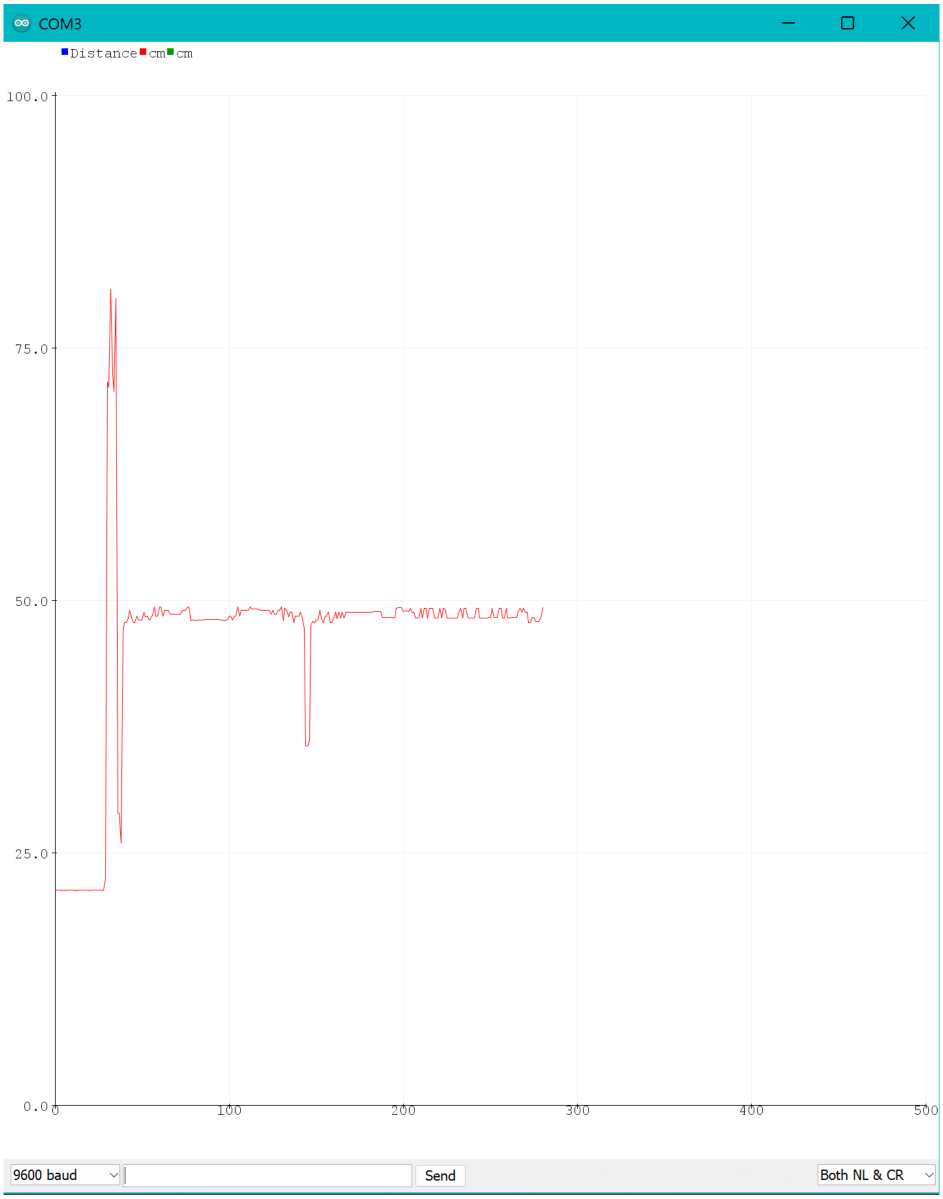

Figure 2: PlotterData

Topic: Section 2

Place your picture of the 74HC595 and LED array. Add a very brief caption (1 sentence) that describes the space savings provided by the 74HC595.

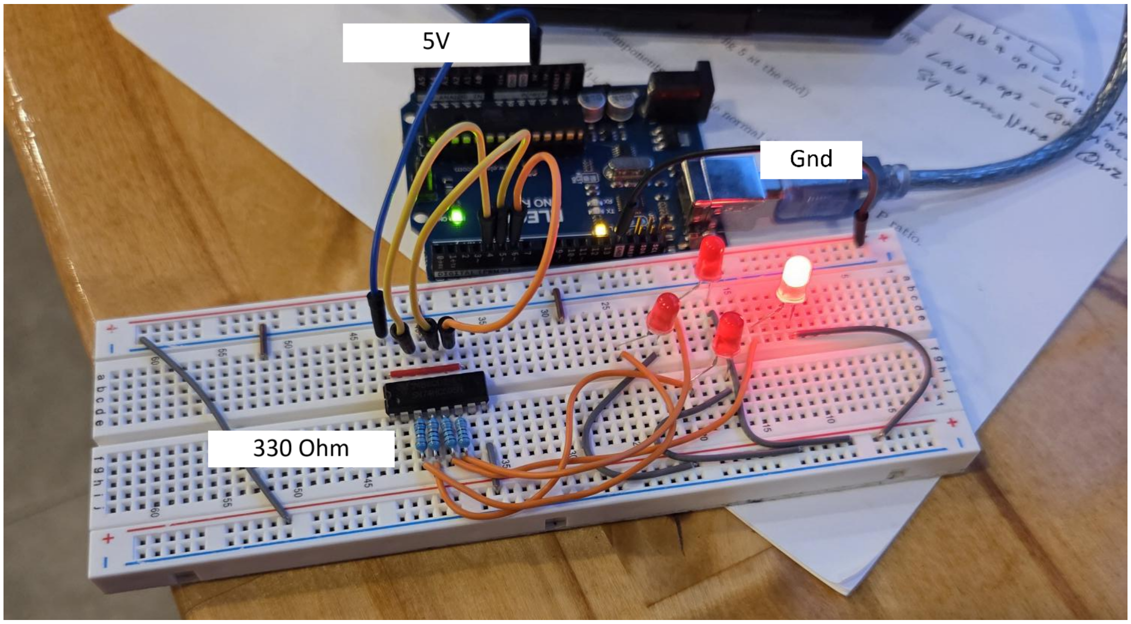

Figure 3: Spinny Lights

Where we would have needed 8 digital pins to drive the 8LED example, the shift register makes it to where we only need 3.

Topic: Section 3
Place your picture of the joystick and LED array. Add a brief caption (1-3 sentences) that describes what the joystick is controlling in the LED array, including direction and strength.

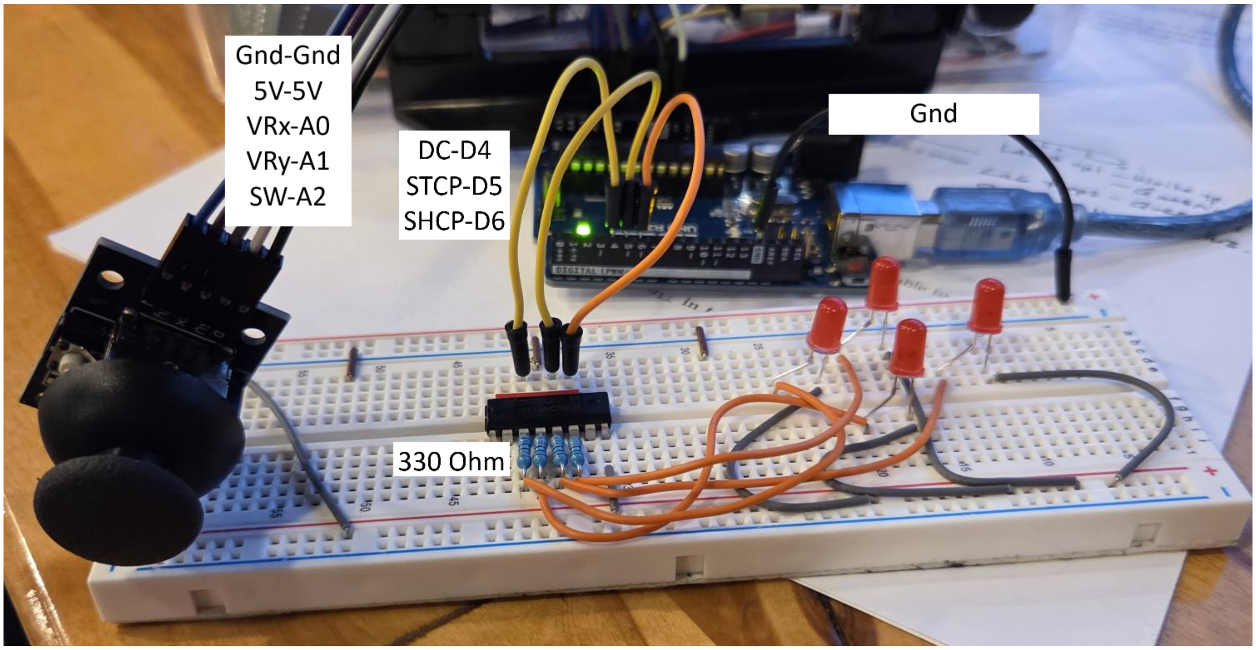

Figure 4: Joy Stick Lights

The joystick contains 2 potentiometers that send signals to the analog input pins. both potentiometers lay centered at mid resistence and moving to or from min or max gives the values for the code to then send the signal to the right pin. The direction is set with how the codes math is set by

int x = centerX − xRaw
int y = yRaw − centerY

Topic: Section 4

Please list the following:
1. Any online sources you used and what they helped with. If you used any generative AI tools (e.g., ChatGPT, Gemini, Copilot, Claude, etc.), include a screenshot of your entire chatlog.
2. Who you worked with in the lab (excluding teachers) and what they helped with.

Online sources should be notated cleanly with the name and a hyperlink. “What they helped with” should focus on conciseness.
Examples: “Wikipedia helped me understand citations,” “John worked with me on Sign-off 2.”
If the answer is “No online sources used.” or “No collaborators.”, please note this in the report for the corresponding item, so we know you considered both prompts.

No other sources outside of the ones linked in the Lab.

I worked with Madison Flanders, she was able to get the wiring figured out on part 2.

Page 5: Self-evaluation:

(a) What did you expect to learn? With how "Mathy" the filtering one seemed I was thinking that this was going to be more intensive. I was hoping to be able to maybe get back on to the liquid LCD again and do the balancing act of hitting a specific voltage and outputting values. This was okay though.

(b) How would you rate your attendance and participation in class during this lab experience? Are any adjustments needed? 3.5/4. I feel about average for my typical involvement with the lab. I think I am just getting tired here at the end of the term.

(c) How well did you do at the lab assignment itself? You might include thinking about strengths and weaknesses of your performance in this response. Please also give yourself a grade (out of 4 possible points). I would give myself a 4 for this. I kept on top of everything in lab. Considering the world is ending, I call that a win.

(d) What did you actually learn in the lab? (This might include a range of things, from subject matter and concrete skills to ways of knowing and working together with peers.) I definitely have a larger appreciation for the work that is done at the bit level. I think that there is this idea in students mind that controls engineering is NOT done that close to the metal but its just not true. Especially when we start to think about truly robust systems controls solutions.

Code

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%CODE HERE
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